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**Question 1.2.** State the null hypothesis. It should reflect the position of either Dr. DeNero or Dr. Sahai.  
(4 points)

*Note:* Check out [11.3](#) for a refresher on hypotheses.

Null Hypothesis:  $H_0$ : The probability of a person taking the Vaccine 1 is 0.6



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**Question 1.3.** State the alternative hypothesis. It should reflect the position of the doctor you did not choose to represent in Question 1.2. (4 points)

*Note:* Check out [11.3](#) for a refresher on hypotheses.

Alternative Hypothesis: H1: The probability of a person receiving Vaccine 1 is not 0.6.



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**Question 1.7.** Simulate 10,000 values of the test statistic under the assumption you picked in Question 1.6. (4 points)

As usual, start by defining a function that simulates one value of the statistic. Your function should use `sample_proportions`. (You may find a variable defined in Question 1.1 useful here!) Then, write a `for` loop to simulate multiple values and collect them in the array `simulated_statistics`.

Use as many lines of code as you need. We have included the code that visualizes the distribution of the simulated values. The red dot represents the observed statistic you found in Question 1.5.

```
In [18]: def one_simulated_statistic():
        proportions = sample_proportions(sample_size, [0.6, 0.4])
        percent_sim = proportions[0] * 100
        return abs(percent_sim - 60)
```

```
In [23]: # Run the this cell a few times to see how the simulated statistic changes
        one_simulated_statistic()
```

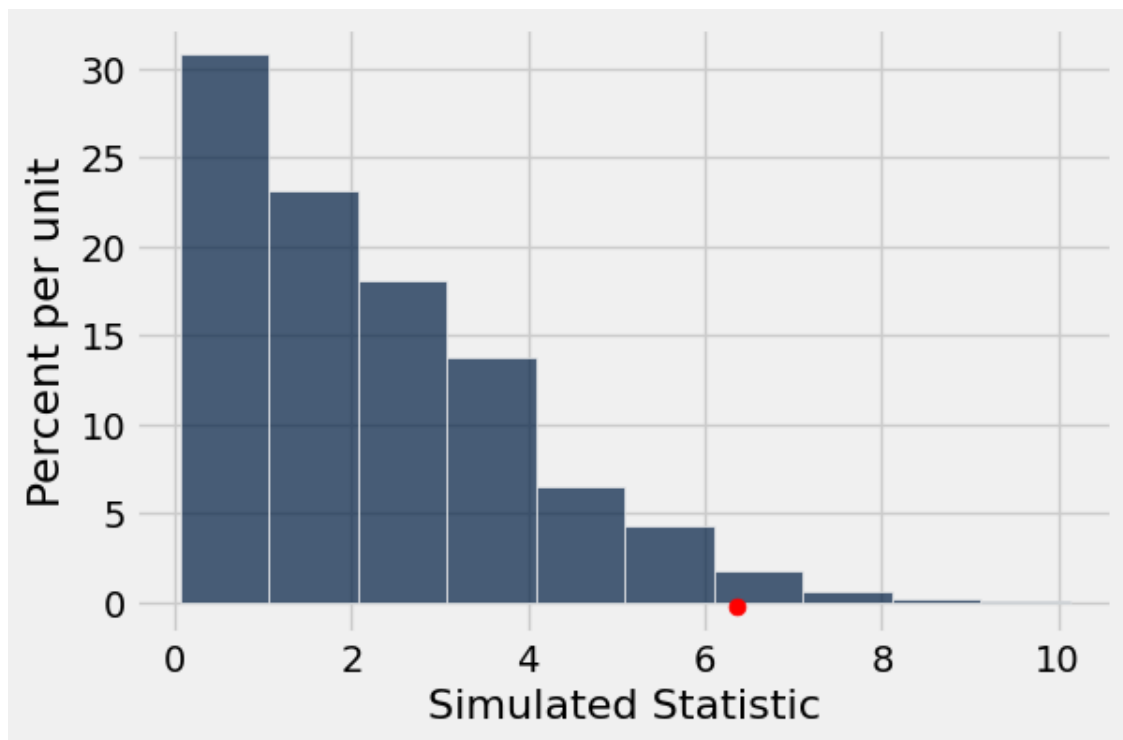
```
Out[23]: 5.4088050314465335
```

```
In [24]: num_simulations = 10000

        simulated_statistics = make_array()
        for i in range(num_simulations):
            simulated_statistics = np.append(simulated_statistics, one_simulated_statistic())
```

```
In [25]: # Run this cell to produce a histogram of the simulated statistics

        Table().with_columns('Simulated Statistic', simulated_statistics).hist()
        plt.scatter(observed_statistic, -0.002, color='red', s=40);
```



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**Question 2.1.** Suppose we want to test whether or not each factor contributes the same amount to the overall Happiness Score. Define the null hypothesis, alternative hypothesis, and test statistic in the cell below. **(4 points)**

*Note:* Please format your answer as follows: - Null Hypothesis: ...  
- Alternative Hypothesis: ...  
- Test Statistic: ...

Null Hypothesis: All factors have the same contribution

Alternative Hypothesis: The contributions of the factors are unequal

Test Statistic: Total Variation Distance (TVD) between the observed and uniform distributions





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**Question 2.5.** What can you conclude about how each factor contributes to the overall happiness score in the US? Explain your answer using the results of your hypothesis test. Assume a p-value cutoff of 5%. (4 points)

*Type your answer here, replacing this text.*

